

MSK Pulse

Project Rationale & Business Alignment

Why this project was built, and how its scope maps to a health-tech operations & analytics function

Why This Project Exists

Health-tech companies operating clinic networks depend on two things working together: analytics good enough to trust for real operating decisions, and models that act on what those analytics find — deployed somewhere a real workflow can use them, not left in a notebook. MSK Pulse was built as a concrete demonstration of exactly that combination, applied to a realistic, high-stakes operational problem: prior-authorization denial in a musculoskeletal (MSK) care setting.

The project was scoped end-to-end on purpose — synthetic data generation, a tested warehouse, a metrics dictionary, five real analyses, a calibrated and explainable model, a live API, and a dashboard a non-technical operator could use — because that full chain is what the work actually looks like in this kind of role, not any single piece of it in isolation.

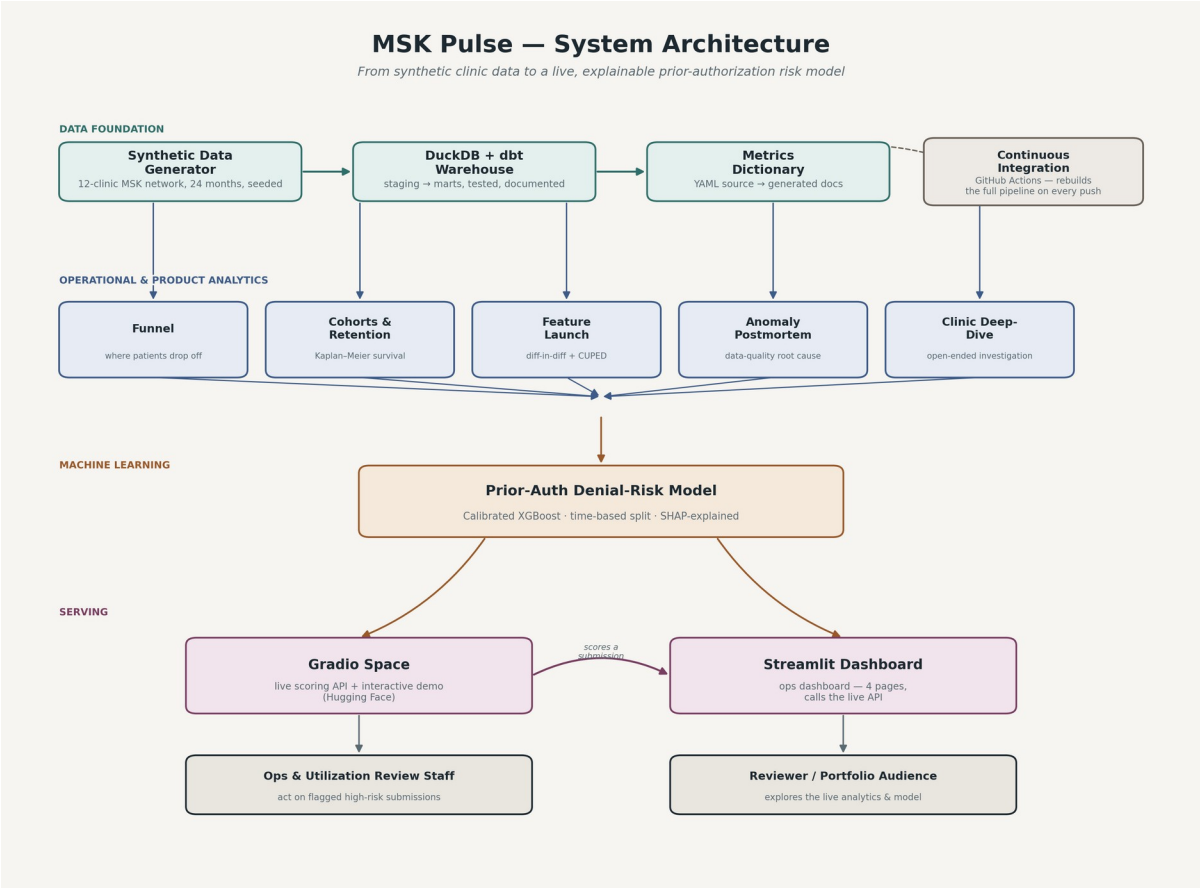
How the Project Maps to the Kind of Work This Represents

Area of responsibility	What MSK Pulse demonstrates
Understanding the business problem before modeling	The funnel analysis quantifies where patients actually drop off first, and shows prior-auth denial is the largest controllable loss point — the model built afterward targets that specific, evidence-backed problem rather than an assumed one.
Trustworthy measurement at scale	A dbt-tested warehouse and a single versioned metrics dictionary mean every number — in an analysis memo, a dashboard, or a model feature — is defined once and traces back to the same source.
Rigorous, not just directional, analysis	Feature-launch impact is measured with difference-in-differences (not a naive before/after), retention with Kaplan–Meier survival curves, and a data-quality break is caught by an automated warehouse test and worked as a full postmortem.
Production ML, not notebook ML	The denial-risk model is calibrated (so a "22% risk" score is a real probability), explained per-prediction with SHAP, served behind a live API, and consumed by an actual dashboard page — the same shape as a model reaching a real operational workflow.
Operating under real platform constraints	Deployment hit genuine constraints (a hosting platform's paid-tier limits, a library version incompatibility, a silent API contract mismatch) and each was diagnosed and resolved rather than avoided — documented in full in the project's technical documentation.

Area of responsibility	What MSK Pulse demonstrates
Communicating findings clearly	Every analysis ships with a written memo aimed at a non-technical stakeholder, in addition to the code that produced it.

System Overview

The project is one pipeline: synthetic clinic data flows into a tested warehouse, which feeds both the operational analyses and the risk model; the model is served live and consumed by an operations-facing dashboard.



System architecture overview.

Summary

MSK Pulse is best read as a working proof of the full cycle this kind of role is responsible for: diagnose an operational problem with rigorous analysis, build the infrastructure needed to trust that diagnosis, and ship a model that acts on it somewhere real — end to end, and honestly documented along the way, including what went wrong and how it was fixed.